PAUL STEPHEN

Software Engineer – Backend & AI/ML

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PROFESSIONAL SUMMARY

Software engineer with 3 years of experience taking ML and GenAI systems from design to production and keeping them reliable. I build backend services, RAG pipelines and ML models that cut manual work, protect SLAs and scale to tens of thousands of daily events. Focus on privacy-first, self-hosted, cost-efficient architecture for regulated, document-heavy industries (finance, insurance).

KEY STRENGTHS

- **Business results:** routing accuracy 71% → 97%, SLA breaches down ~80%, document search from ~20 min to under 10 sec.
- **Production ownership:** design, ship, monitor and maintain, with fallbacks, caching, rate limiting, retries and observability.
- **Practical AI tooling:** local LLMs, embeddings and retrieval to automate triage, search and analysis without sending data to third parties or paying per-call API costs.
- **Full-stack reach:** backend, ML, data pipelines and Flutter/React front ends, delivering features end to end.

PROFESSIONAL EXPERIENCE

AI Backend Engineer – Kinara Capital, Bengaluru *Feb 2024 – Present*

AI Ticket Intelligence (ML / NLP)

- Built embedding-based ticket routing: accuracy **71% → 97%**, cutting manual triage across 340+ tickets/month.
- Built SLA-risk prediction with early escalation signals: **~80% fewer SLA breaches**, with feedback monitoring and rule-based fallbacks for production safety.

Enterprise RAG Document Platform (GenAI)

- Designed and shipped a fully self-hosted RAG pipeline (ingestion, OCR/parsing, chunking, embedding, hybrid retrieval, grounded Q&A) on **FAISS** and **Ollama**, with zero external API cost.
- Cut document search from **~20 min to under 10 sec**; reached **91% answer relevance** through hybrid retrieval and chunking tuning; added caching, rate limiting and async job orchestration.

High-Throughput Async Backend

- Re-architected a synchronous pipeline into Python asyncio services: **5x throughput** (200 → 1,200+ emails/min), 50,000+ events/day.
- Added retries, exponential backoff and dead-letter queues: delivery reliability **95.8% → 99.7%**.

AI-Enabled Loan Platform (Full-Stack)

- Built Flutter + FastAPI workflows (auth, APIs, data pipelines) for **10,000+ users** with zero critical incidents in 12 months; shipped AI repayment insights into the product.

SELECTED PROJECTS

- **Agentic Codebase Analyzer (LangGraph, LangChain, Ollama, FAISS):** local multi-stage agent that maps dependencies and generates architecture summaries.
- **Smart URL Shortener (FastAPI, pgvector, Redis, PostgreSQL):** semantic duplicate/safety detection and popularity prediction with caching.
- **MoneyFlowX (Flutter, ML):** offline transaction categorization, anomaly detection and budget forecasting.

TECHNICAL SKILLS

- **Backend:** Python, FastAPI, Node.js, Asyncio, REST, Microservices, WebSockets, Distributed Systems
- **AI / ML:** RAG, LLM inference (Ollama), LangChain, LangGraph, Embeddings, FAISS, pgvector, NLP, PyTorch, Evaluation & Guardrails
- **Data & Cloud:** PostgreSQL, MongoDB, Redis, AWS (Lambda, EC2, S3), Docker, Nginx, Linux, GitHub Actions CI/CD
- **Frontend:** Flutter (iOS/Android/Web), React

EDUCATION & CERTIFICATIONS

- **Bachelor of Computer Applications (BCA)** – Kristu Jayanti College, Bengaluru (2021–2024)
- IBM Full Stack Development Certification
- AWS Cloud Technical Training